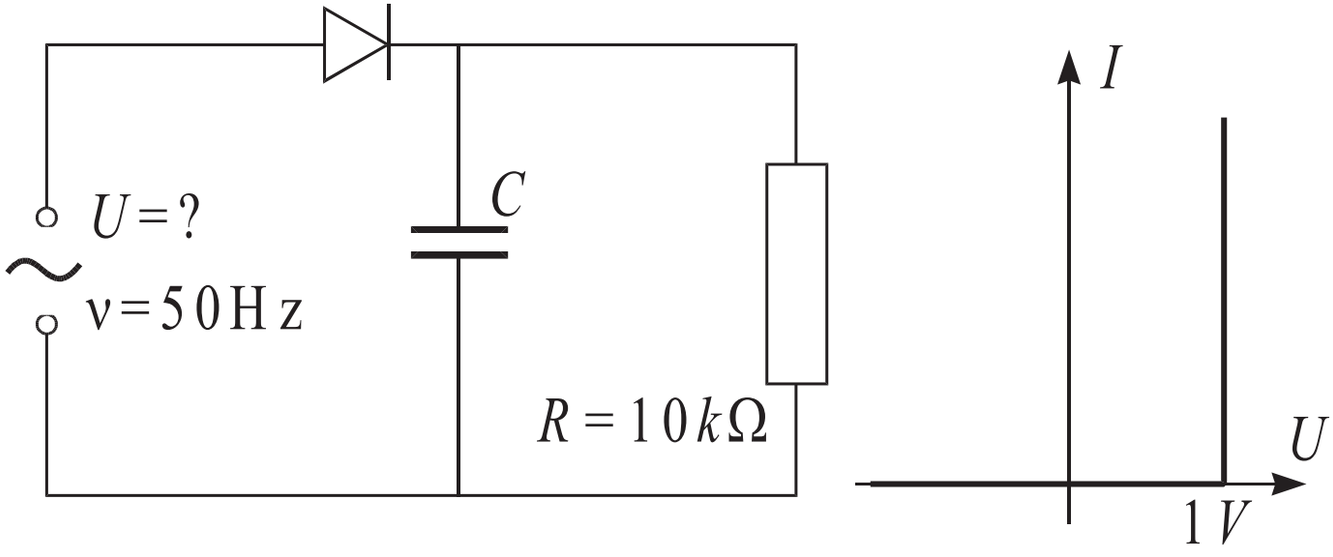
6. **Rectifier** (8 points)

A voltage rectifier is made according to the circuit depicted in Figure. The load $R = 10\,\mathrm{k\Omega}$ is fed with DC, equal to $I = 2\,\mathrm{mA}$. In what follows we approximate the U-I characteristic of the diode with the curve depicted in Figure. The relative variation of the current at the load has to satisfy the condition $\Delta I/I < 1\%$.



1) Find the average power dissipation at the diode at the working regime of such a circuit.

2) Determine the amplitude of the AC voltage (with frequency $\nu = 50\,\mathrm{Hz}$), which has to be applied at the input of the circuit.

3) Find the required capacitance C.

4) Find the average power dissipation at the diode during the first period (of AC input voltage) immediately following the application of AC voltage to the input of the circuit.